

EDUCATION

University College London

PhD in Statistical Science

London, UK

2022–Present

- Topic: Robust Bayesian methods with conjugacy properties
- Supervisors: Dr. Jeremias Knoblauch, Dr. François-Xavier Briol

Universidad de Chile

M.S. in Engineering Sciences, Applied Mathematics

Santiago, Chile

2020–2021

- Thesis: “Nonstationary multi-output Gaussian processes via harmonizable spectral mixtures”
- Supervisor: Dr. Felipe Tobar
- Ranked in top 10%, best thesis award

École Centrale Paris

Student Exchange Program

Paris, France

Spring 2017

Universidad de Chile

B.S. in Engineering Sciences, Mathematics

Santiago, Chile

2014–2019

- Ranked in top 1%, 2 awards for outstanding academic achievement

PUBLICATIONS

- [1] W. Laplante, **M. Altamirano**, A. Duncan, J. Knoblauch, and F.-X. Briol, “Robust and conjugate spatio-temporal gaussian processes”, *arXiv preprint arXiv:2502.02450*, 2025, To appear at ICML 2025.
- [2] **M. Altamirano**, F.-X. Briol, and J. Knoblauch, “Robust and conjugate Gaussian process regression”, in *International Conference on Machine Learning*, spotlight, 2024, pp. 1155–1185.
- [3] G. Durán-Martín, **M. Altamirano**, A. Y. Briol Shestopaloff, L. Sánchez-Betancourt, J. Knoblauch, M. Jones, F.-X. Briol, and K. Murphy, “Outlier-robust Kalman filtering through generalised bayes”, in *International Conference on Machine Learning*, 2024, pp. 12 138–12 171.
- [4] **M. Altamirano**, F.-X. Briol, and J. Knoblauch, “Robust and scalable Bayesian online changepoint detection”, in *International Conference on Machine Learning*, 2023, pp. 642–663.
- [5] **M. Altamirano** and F. Tobar, “Nonstationary multi-output Gaussian processes via harmonizable spectral mixtures”, in *International Conference on Artificial Intelligence and Statistics*, 2022, pp. 3204–3218.
- [6] **M. Altamirano**, P. Uribe, D. Schlotterbeck, A. Jiménez, R. Araya, J. van der Molen Moris, and D. Caballero, “Unsupervised characterization of lessons according to temporal patterns of teacher talk via topic modeling”, *Neurocomputing*, 2021, ISSN: 0925-2312.
- [7] **M. Altamirano**, A. Jiménez, and R. Araya, “Lessons clustering using topics inferred by unsupervised modeling from textbooks”, in *International Conference in Methodologies and intelligent Systems for Technology Enhanced Learning*, Springer, 2020, pp. 85–94.

EXPERIENCE

Bloomberg L.P.

Research Intern

London, UK

July 2025–September 2025

Matterhorn Studio

Research Scientist

London, UK

January 2024–September 2024

Universidad de Chile - Center for Mathematical Modeling

Research Assistant, Supervisor: Dr. Joaquín Fontbona

Santiago, Chile

September 2021–June 2022

Universidad de Chile - Center for Advanced Research in Education

Research Assistant, Supervisor: Dr. Roberto Araya

Santiago, Chile

March 2019–January 2022

- Designed and implemented unsupervised learning models to analyse temporal patterns of teacher talk.
- Published findings in reputable journals and conferences as an undergraduate: [6], [7]

AWARDS & GRANTS

- **Prob_AI Hub - Networking and Development Fund** worth £9950 to organise workshop in post-Bayesian methods. 2025
- **Bloomberg Data Science Ph.D. Fellowship.** 2024–Present
- **UCL Costas Goutis Prize.** Awarded by the Department of Statistical Science for the best PhD progress. 2024
- **International Conference on Machine Learning (ICML) Spotlight [2].** Awarded to 3.5% out of more than 9,000 submissions as a marker of quality and significance. 2024
- **Poster prize winner** worth £500 judged by Prof. Guy Nason, UCL Departmental poster session. 2023
- **UCL Departmental Teaching Assistant studentship** for 3.5 years worth around £150,000. 2022
- **Best thesis**, Msc in applied mathematics, Universidad de Chile. 2022
- **Best Paper Award** at 10th International Conference in Methodologies and Intelligent Systems for Technology Enhanced Learning [7]. Achieved as an undergraduate student. 2020
- **Center of Mathematical Modelling (CMM), Msc in applied mathematics Studentship** worth around £10,000. 2020
- **Outstanding student**, Faculty of Physical and Mathematical Sciences, Universidad de Chile. 2015/2019

TALKS & POSTERS

Talks:

• **Robust and Conjugate Gaussian Process Regression**

- Satellite Workshop BayesComp 2025, Singapore, Singapore June 2025
- International Conference on Robust Statistics (ICORS), Stresa, Italy May 2025
- Imperial College London CSML reading group, London, UK February 2025
- Department of Computer Science and Technology, Cambridge University, Cambridge, UK February 2025
- ICML London Meetup, London, UK July 2024

• **Robust and Scalable Bayesian Inference**

- Bayesian Young Statisticians Meeting, Online April 2025
- Newcastle University Statistics Seminar, Newcastle, UK May 2024
- UCL Statistical Science PhD Seminars, London, UK May 2024
- UCL Department of Electronic and Electrical Engineering, London, UK March 2024

- **Robust and Scalable Bayesian Online Changepoint Detection**
 - Oxford-Man Institute Seminar: Statistical Machine Learning and Finance, Oxford, UK December 2024
 - UCL Distance-Based Methods in Machine Learning Workshop, London, UK June 2023

Posters:

- **Robust and Conjugate Gaussian Process Regression**
 - BayesComp 2025, Singapore, Singapore June 2025
 - International Conference on Machine Learning (ICML), Vienna, Austria July 2024
 - Symposium on Advances in Approximate Bayesian Inference (AABI), Vienna, Austria July 2024
 - Workshop on Functional Inference and Machine Intelligence (FIMI), Bristol, UK March 2024
 - UCL Statistical Science Departmental Poster Session, London, UK Nov 2023
- **Robust and Scalable Bayesian Online Changepoint Detection**
 - International Conference on Machine Learning (ICML), Hawaii, US July 2023
 - Symposium on Advances in Approximate Bayesian Inference (AABI), Hawaii, US July 2023
 - SIAM UKIE National Student Chapter Conference, Oxford, UK June 2023
 - London Symposium on Information Theory, London, UK May 2023

SERVICES TO THE COMMUNITY

- **Reviewing activities:** AISTATS (2023, 2024, 2025), NeurIPS (2023, 2024), ICLR (2025), ICML (2023 Workshop on Structured Probabilistic Inference & Generative Modeling, 2025), Statistics and Computing, Bayesian Analysis
- **Organisation of Workshops & Events:** Co-organiser of [Advances in post-Bayesian methods workshop](#), Co-organiser of [Post-Bayes seminar series](#), Co-organiser of [UCL Distance-based methods in Machine Learning workshop](#), co-organiser of asymptotic statistics reading group, UCL.

TEACHING

- **Teaching Assistant** at UCL 2023
STAT0004 (Term 1 2022/Term 2 2022/Term 2 2023), STAT0005 (Term 1 2022/Term 1 2023), STAT0023 (Term 2 2022), STAT0040 (Term 1 2023/ Term 1 2024)
- **Teaching Assistant** at Universidad de Chile 2016-2020
Multivariable Calculus (Fall 2016/Fall 2018/Fall 2019/Spring 2019/Fall 2020), Advanced Calculus and Applications (Spring 2019), Linear Algebra (Spring 2017/Spring 2018), Introduction to Calculus (Spring 2017)

CODING AND COMPUTER SKILLS

- **Languages:** Python, Matlab, Java, SQL, R
- **Additional Skills:** Linux, $\text{\LaTeX}$, Github

LANGUAGES

- **Spanish:** Native Speaker
- **English:** Fluent
- **French:** Basic